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RESEARCH ARTICLE

Construction, characterization and application of a genome-wide promoter library in *Saccharomyces cerevisiae*

Ting Yuan[1*], Yakun Guo[1*], Junkai Dong[1], Tianyi Li[1], Tong Zhou[1], Kaiwen Sun[1], Mei Zhang[2], Qingyu Wu[1], Zhen Xie[3], Yizhi Cai[4], Limin Cao[2], Junbiao Dai (✉)[1]

1 MOE Key Laboratory of Bioinformatics and Centre for Synthetic and Systems Biology, School of Life Sciences, Tsinghua University, Beijing 100084, China
2 College of Life Sciences, Capital Normal University, Beijing 100048, China
3 MOE Key Laboratory of Bioinformatics and Bioinformatics Division, Center for Synthetic and Systems Biology, TNLIST/Department of Automation, Tsinghua University, Beijing 100084, China
4 School of Biological Sciences, The King's Buildings, University of Edinburgh, Edinburgh, EH9 3BF, UK

Abstract Promoters are critical elements to control gene expression but could behave differently under various growth conditions. Here we report the construction of a genome-wide promoter library, in which each native promoter in *Saccharomyces cerevisiae* was cloned upstream of a yellow fluorescent protein (YFP) reporter gene. Nine libraries were arbitrarily defined and assembled in bacteria. The resulting pools of promoters could be prepared and transformed into a yeast strain either as centromeric plasmids or integrated into a genomic locus upon enzymatic treatment. Using fluorescence activated cell sorting, we classified the yeast strains based on YFP fluorescence intensity and arbitrarily divided the entire library into 12 bins, representing weak to strong promoters. Several strong promoters were identified from the most active bins and their activities were assayed under different growth conditions. Finally, these promoters were applied to drive the expression of genes in xylose utilization to improve fermentation efficiency. Together, this library could provide a quick solution to identify and utilize desired promoters under user-defined growth conditions.

Keywords synthetic biology, yeast, promoter activity, metabolic engineering, xylose utilization

1 Introduction

Precise regulation of gene expression is pivotal to engineer a metabolic pathway [1–3], in which the promoter serves as an important regulatory element. Recent development in synthetic biology allows assembly of large DNA fragments to become much more convenient [4–6]. In addition, methods on modular design and construction of biological parts were invented and widely applied during the process of constructing a metabolic pathway [6,7]. These techniques make it more convenient to optimize the regulatory network of metabolic pathways and to increase the yield of a target product.

The budding yeast, *Saccharomyces cerevisiae*, has been used as a preferred host for bioethanol production for a long period of time [8]. Pretreated lignocellulose has been used for bioethanol production in industry to reduce the cost [9]. During the fermentation process, many factors could influence the growth and productivity of the yeast, such as acidic matters in the lignocellulose hydrolysates and high ethanol concentration. Therefore, much work has been done to try to solve these problems [10–12]. According to the change gene regulation network during fermentation, the yield and productivity of ethanol could be improved by using specific promoters to regulate the metabolic flux [13].

Characterizing biological parts has been done extensively in *Escherichia coli*. For examples, hundreds of parts have been constructed and profiled as BioBricks [14,15]. The number of parts for higher eukaryotic organisms such as *S. cerevisiae* is still very limited. Previously, we constructed a lot of promoters in budding yeast as standard parts, and measured their activity under normal growth condition (yeast extract-peptone-dextrose, YPD) or using some stress treatments such as oxidative stress [7]. However, constructing and measuring the activity of promoters individually could be time-consuming and

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E-mail: jbdai@tsinghua.edu.cn

*These authors contributed to this work equally laborious. In previous study, Sharon and colleagues developed a high-throughput method to classify the synthetic promoters using fluorescence-activated cell sorting (FACS) analysis [16]. In this study, we constructed a library of reporter plasmids, covering every native promoter within the yeast genome, and profiled the promoters based on the fluorescence intensity to define their activity. The pooled strains allow us to quickly identify the promoters, which remain active under various conditions by FACS analysis. Finally, using selected promoters, we constructed the xylose metabolic pathway and demonstrated that the strains by controlling genes using selected strong promoters utilize xylose better than those using weak promoters.

2 Materials and methods

2.1 Strains, plasmids and growth conditions

E. coli DH5α was used for plasmid construction and amplification. Yeast strain BY4741 (MATa *leu2Δ0 met15Δ0 ura3Δ0 his3Δ1*) was used to host the reporter constructs and subjected to FACS analysis. Yeast strain JDY52-URR-His3 (MATa *his3Δ200 leu2Δ0 lys2Δ0 trp1Δ63 ura3Δ0 met15Δ0*), a derivative of S288C with I-SceI [17,18] recognition sites flanking the URR1-His3-URR2 fragment at HO locus was used as a host for xylose-utilization pathway. The reporter plasmid pYG026 [7] was used to construct the promoter pool.

Yeast cells were cultured in YPD (10 g/L yeast extract, 20 g/L peptone, 20 g/L glucose) medium, otherwise cells were cultured in YPDX (10 g/L yeast extract, 20 g/L peptone, 20 g/L glucose, 20 g/L xylose) medium for fermentation.

2.2 Construction of promoter pools

Primers used to amplify promoters were commercially synthesized and distributed in 96-well plates. DNA polymerase KOD (TOYOBO) was used for polymerase chain reaction (PCR) amplification to ensure fidelity. PCR program was as follows: 94 °C for 3 min, 10 cycles of 94 °C 15 s, 48 °C 30 s, 68 °C 30 s, 10 cycles of 94 °C 15 s, 52 °C 30 s, 68 °C 30 s, 10 cycles of 94 °C 15 s, 55 °C 30 s, 68 °C 30 s, 68 °C for 5 min. For each sub-library (library 1–9), 1 μL of PCR products from each well of 96-well plates were mixed together. The mixture was column-purified and assembled into the reporter plasmid (in 100 μL total reaction volume): 10 μL 10 X T4 ligase buffer (Thermo Scientific), 1 μL 100 X BSA (NEB), 250 ng reporter plasmids, 50 μL purified PCR product, 40 U of BsaI (NEB), 5 U T4 ligase (Thermo Scientific) and appropriate amount of ddH_2O. The above reaction mixture (20 μL) was added to one PCR tube and incubated using the following program: 37 °C for 1 h, 50 °C for 15 min, 80 °C for 15 min and keep at 10 °C. After incubation, the mixtures were transformed into DH5α and resistant clones were selected on LB medium containing 50 μg/mL carbenicilin (Inalco S.p.A.). At the same time, a 1 : 100 cells suspension was plated onto a separate plate to count colony number and confirm randomly picked clones by colony PCR using primers YGO199 (5′-gcgtatatataccaatc-taagtct-3′) and YGO200 (5′-gtcaatttaccgtaagtagcatc-3′). The assembly efficiency was calculated. Eventually, two larger libraries (library 1–4, library 5–9) of bacteria strains were generated and the plasmids were isolated.

2.3 Integrate the reporter construct into yeast genome

The plasmids were digested overnight with BciVI (NEB), transformed into BY4741 and selected on synthetic complete medium lacking uracil (SC-Ura) plate. The number of transformants was counted and the integration efficiency was calculated based on PCR analysis with primer YGO020 (5′-acgaaaagagagatcacatggtcttgttaga-3′) and YGO309 (5′-cacgaacttctatatgctcgc-3′), YGO027 (5′-gtgatcgtcacaaaccgtaag-3′) and YGO246 (5′-cagtgatggga-gacggcatgtctaaaggtgaagaattattcac-3′). The depth of the yeast library was calculated as follows: Depth = the total number of yeast colony × integration efficiency/total number of promoters within this library. Then the pooled yeast strains were collected and stored at –80 °C.

2.4 Stress treatment and serial dilution analysis

For stress treatment, an aliquot of pooled yeast cells was firstly cultured overnight and transferred to fresh YPD medium with starting OD_{600} at 0.1. After the cells were cultured for 8 h, 100% acetic acid or ethanol was added into the media to reach the desired stress concentration. The cells were treated for 3 h before they were collected, washed with sterile ddH_2O and resuspended in 1 mL sterile ddH_2O. 0.02 OD cells were transferred into 100 μL of ddH_2O to obtain the first dilution, followed by 10-fold serial dilution. The cell suspension (3 μL) was spotted onto selective plates and incubated at 30 °C for 1 d before the pictures were taken.

2.5 FACS analysis

An aliquot of pooled yeast cells was thawed and inoculated into fresh YPD medium with starting OD_{600} at 0.1. Cells were cultured at 30 °C for 8 h and divided as control sample, which was collected for FACS analysis directly, and stress-treated samples, which were cultured in medium containing 10% ethanol (Beijing Chemical Works) or 0.8% acetate acid (Beijing Chemical Works) for another 3 h before FACS analysis.

Cells were washed twice with sterile ddH_2O and resuspended in SC-Ura medium before FACS analysis. The Arial II cell sorter (BD FACSAria SORP.) was used to sort the cells. The 488 nm laser and 530/30 nm filter were used for detection of YFP fluorescence intensity, whereas the 561 nm laser and 610/20 nm filter for mCherry fluorescence. Only the mCherry fluorescence positive cells were selected and then gated to 12 different bins according to YFP fluorescence/mCherry fluorescence.

2.6 Multiplex sequencing library preparation

Cells from each bin were cultured and their genomic DNA was extracted. Promoter regions were amplified with KOD plus DNA polymerase (TOYOBO). Forward primer 5′-acactctttccctacacgacgctcttccgatcttgaggatcccagtggct-3′ and reverse primer 5′-tgactggagttcagacgtgtgctcttccgatctCTTCACCTTTAGACATaaaaacCATc-3′ were used for first round PCR with the following program: 94 °C 3 min; 30 cycles of 94 °C 15 s, 54 °C 30 s, 68 °C 35 s; 68 °C 5 min. PCR products were purified, diluted (1 : 100) and used as template for the second round PCR. Forward primer 5′-aatgatacggcgaccaccgagatctacactctttccctacacgacg-3′ and reverse primer 5′-caagcagaagacggcatacgagatNNNNNNNNgtgactggagttcagacgtgtgctc-3′ (NNNNNNNN represents the sequencing barcode of each pool) were used in the second round PCR. The following program was used: 94 °C 3 min; 20 cycles of 94 °C 15 s, 52 °C 30 s, 68 °C 40 s; 10 cycles of 94 °C 15 s, 58 °C 30 s, 68 °C 40 s; 68 °C 5 min. PCR products were purified and subjected to sequencing.

2.7 Analysis of sequencing data

The sequencing reads were assigned to each sample according to the 8-bp index. Then the reads were mapped to the genome to find the promoter ID. We first normalized the reads of promoters in each bin to cell numbers of promoters in that bin (n_i, i represents the bin number 1–12) per 10^7 total library cells, and n_i is defined as $A/N \times 10^7 \times a\%$, where A is the number of sequencing reads of a promoter, N is the total number of reads in a bin, 10^7 is the total number of cells per library and $a\%$ is the fraction of the bin in the whole library. Then for a certain promoter X, its activity is defined as $\Sigma P_i \times n_i/Nx$, where i is bin number (from 1 to 12), P_i is the mean promoter activity of bin i, and Nx is total cell number of promoter X in all bins (the total number of n_1 to n_{12}).

2.8 Measurement of individual promoter activity

For promoter activity, which was measured individually, cells were cultured in YPD medium at starting $OD_{600} = 0.1$ for 8 h, pelleted, washed with PBST, and resuspended in PBST before analysis following protocol described previously [7].

2.9 Construction of yeast strains that could utilize xylose

For the construction of yeast strains (YTY1, YTY2, YTY3, YTY4), we first assembled transcription units (TUs) (Table 1). The identity of the promoter (pADH1, pPGK1, pCYC1, pYHR135C, pYHR021C, pYIL069C, pYKL060C and pYHR174W) was determined using the Basic Local Alignment Search Tool (BLAST) at Saccharomyces Genome Database (SGD) (http://www.yeastgenome.org/). Then we assembled these transcription units into metabolic pathways. All the methods we followed were described in a previous paper [7].

2.10 Ethanol fermentation assay

The yeast strains were first cultured in 15 mL YPD medium overnight at 30 °C. Cells were centrifuged at 3000 r/min for 5 min at 4 °C, washed twice with sterile water, transferred to 250 mL conical flasks, each of which contains 100 mL YPDX medium composed of approximately 20 g/L xylose (Sigma) and 20 g/L glucose (Sigma) at an initial cell density of $OD_{600} = 0.6$, and then incubated at 30 °C with 200 r/min shaking. All flasks were covered with one layer of tinfoil immediately after inoculation. Samples (1 mL) were taken at the indicated time intervals [13].

2.11 Analysis of metabolites during xylose fermentation

The samples were centrifuged at 12000 r/min at 4 °C, then 100 µL supernatant of each sample was transferred into 900 µL sterile water and filtered through the 0.22 um filter membrane. And then glucose, xylose, xylitol and ethanol were analyzed by HPLC (Agilent Technologies 1260 Series USA) using an Aminex HPX 87H column (Bio-Rad, USA) with a mobile phase of 5 mmol/L H_2SO_4 (Beijing Chemical Works) and a refractive-index detector. The flow rate was 0.4 mL/min, and the column and detection temperatures were 30 °C and 50 °C, respectively [13].

Table 1 TUs of constructed strains

Strains	Transcription units (TUs)	Promoter activity
YTY1	pPGK1-XI-tCYC1, pADH1-XK-tTEF2	Common
YTY2	pYHR135C-XI-tCYC1, pCYC1-XK-tTEF2	Weak
YTY3	pYHR020C-XI-tCYC1, pYIL069C-XK-tTEF2	Medium
YTY4	pYKL060C-XI-tCYC1, pYHR174W-XK-tTEF2	Strong

3 Results and discussion

3.1 Construction of a plasmid library including almost all promoters within the yeast genome

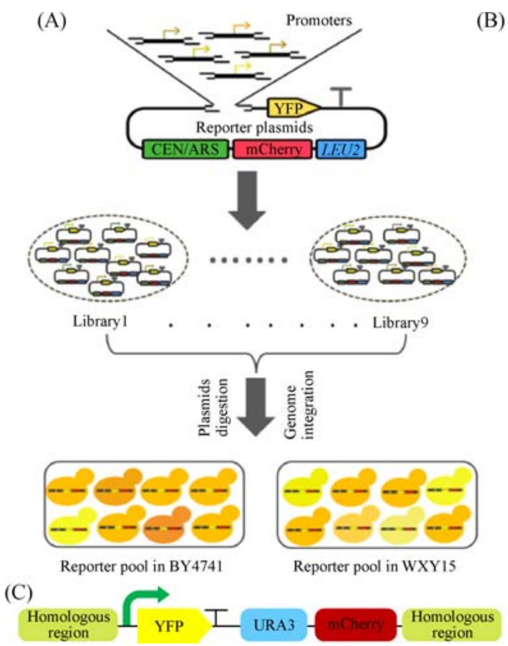
Previously, we individually constructed over 1000 promoters into a designed vector [7]. These promoters could be subcloned into a reporter vector to measure the activity respectively. Using this method, we profiled the activity of about 200 promoters. However, it would be very laborious and time-consuming to finish testing all of the near 6000 promoters.

To overcome this problem and expedite the measuring process, an alternative approach was adopted. As shown in Fig. 1(A), we arbitrarily divided almost all of the promoters into nine separate libraries and each generated a pool of plasmids, in which each promoter was placed upstream of the YFP reporter in the same vector as described before [7]. Next, these plasmids were isolated from each library, pooled together, digested with restriction enzyme, and transformed into the lab strain BY4741 or the industrial strain WXY15 [13].

To obtain the libraries, each promoter was amplified using a pair of specific primers in 96-well plates. A total of 413 to 864 PCR reactions were arbitrarily pooled to form each library, assembled into the reporter vector following the one-step Yeastfab method [7] and transformed into bacteria (Fig. 1(A)). For each library, at least 24 clones were randomly isolated and subjected to colony PCR to confirm the presence of the inserted promoter within the reporter plasmid. We found over 70% of the clones were correctly assembled, suggesting that the Yeastfab assembly method is very efficient. In addition, to determine the coverage of each library, transformants were counted respectively. As shown in Fig. 1(B), at least 10× depth was obtained for each library, with most of them more than 30× (minimal 13.8× and maxim 440×), suggesting most of the libraries are sufficiently complete.

The nine libraries were subsequently mixed, digested with restriction enzyme to release the DNA fragment which contains the promoter, reporter gene, a selective marker and the flanking homologous sequences (Fig. 1(C)), and transformed into yeast, allowing the DNA fragment to be integrated at the HO locus. Clones were selected, pooled and eventually formed two yeast libraries for both BY4741 and WXY15 (Table 2). Similarly, randomly selected yeast clones were isolated to verify the correct integration of the reporter construct, which revealed over 20× depth for each library.

In summary, we have constructed libraries of bacteria and yeast to include whole genome-wide promoters cloned upstream of a YFP reporter, which could allow us quickly assay the promoter activity under not only normal growth condition but also many other conditions. To obtain



(B)

Bacterial library	Promoters from	Number of promoters	Number of transformants	Assembly efficiency /%	Depth
Library 1	chrIII, chrVI, chrVII, chrVIII, chrIX	864	47000	68.75	37.4
Library 2	chrVII, chrVIII, chrIX	520	30700	84.38	50
Library 3	chrXV, chrVIII	672	35900	87.5	46.7
Library 4	chrXVI	516	~7100	100	13.8
Library 5	chrI, chrX, chrXI	672	247500	83.2	50
Library 6	chrXI, chrXIII	422	8280	87.5	17.2
Library 7	chrXIII, chrXIV	413	19740	83.3	29.7
Library 8	chrIV, chrII, chrV	672	306400	96.4	440
Library 9	chrII, chrIV	864	49700	96.87	55.7

Fig. 1 Construct genome-wide promoter libraries. (A) The workflow to construct both the bacteria and yeast libraries. Every native promoter was amplified by PCR, arbitrarily divided into 9 pools and cloned into the reporter plasmid to drive the expression of YFP gene. Plasmids were prepared from the bacteria pools, mixed together and integrated into the yeast genome at HO locus after enzymatic digestion. (B) The composition and coverage of the 9 bacteria libraries. The assembly efficiency is calculated as the percentage of clones, which could generate a fragment at expected size after colony PCR. At least 24 clones were randomly isolated and analyzed. The library depth equals to the number of transformants divided by the number of promoters and multiply the assembly efficiency. (C) A schematic representation of the DNA fragments to be integrated. Homologous regions are DNA sequences flanking the native HO gene. The arrow represents the promoter and T represents the terminator

Table 2 Depth of yeast reporter library

Bacterial library	Promoter number	BY4741 integration pools depth	WXY15 integration pools depth
Library 1–4	2572	27.8	27.3
Library 5–9	3043	> 50	24

promoters with different strength, researchers often induced random mutation within native promoters [5,19]. The homology of the promoters might cause problems in pathway assembly and integration [7]. Our library covered maximum native promoters on yeast genome and allowed efficient screening under various growth conditions [20]. These libraries would be a resource of great value for the entire community.

3.2 Profile the activity of each promoter using FACS followed by high-throughput sequencing

One important application for budding yeast is in the industry of bioethanol fermentation, which requires the strain to tolerate high concentration of ethanol and sometimes various acids if treated cellulose is used as carbon source [11,12]. Many of the strains are very promising under laboratory fermentation condition, but fail during industrial fermentation. One possible reason is that the two fermentation conditions are very different, which could dramatically inhibit the expression of key enzymes. We hypothesize that the use of promoters, which function equally active in industrial condition, could maintain the expression of these key genes, therefore, leading to much more consistent results. To identify these promoters, one can either perform whole-genome transcriptome analysis using high-throughput sequencing technology [13] of samples prepared during industrial fermentation or, as shown below, profiling the yeast library containing the reporter gene under the control of a given promoter under such condition by FACS.

Two common growth conditions, YPD medium containing ethanol and YPD medium containing acetic acid, were applied to identify highly active promoters in both media. We at first performed serial dilutions and growth test to find a suitable condition under which the cells were stressed but could survive. These conditions would enable us to obtain the yeast strains in which the promoters are activated. As shown in Fig. 2(A), cell growth is largely inhibited when they are plated in media containing 1% acetic acid and 12% ethanol in BY4741. Therefore, we used 0.8% acetic acid and 10% of ethanol in our subsequently experiment. Similar conditions were identified for the industrial strain WXY15. Before sorting, the yeast library was inoculated into fresh media and cultured to log phase, and then switched to the stress condition. Cells were harvested, washed and resuspended before subjecting to FACS analysis (Fig. 2(B)). Based on the ratio of YFP fluorescence/mCherry fluorescence (Fig. 2(C)), which represents the activity of a given promoter, cells were sorted into 12 different bins with Bin 1 as the weakest promoter and Bin 12 as the strongest ones. As shown in Fig. 2(D), the majority of cells had quite small fluorescence ratio, indicating that the promoters in these cells were relatively weak when compared to the known CYC1 promoter (the dash line in Fig. 2(D)). On the other hand, a small fraction of cells show very strong activity. Consistently, we found, during FACS analysis, the number of cells in Bin 12 is not easy to get and requires much longer time to collect the same amount of cells.

Next, we cultured the sorted cells from each bin and tested if it contains a population of cells with promoters at different strength. The cells with the positive mCherry fluorescence were firstly selected, and then measured for YFP fluorescence. As shown in Fig. 2(E), the YFP fluorescence increases, as expected, from Bin 1 to Bin 12, indicating that the sorting strategy can be used to separate the cells based on their promoter activity.

To identify promoters in each sorted fraction, we performed high-throughput sequencing (HTS) by multiplexing samples from all bins, which enable us to not only distinguish the promoters but also quantify the activity of each promoter based on the number of reads. We defined the promoter activity according to the ratio of its reads in each bin, the relative mean activity of each bin (normalized to CYC1) and the fraction of each bin in the whole library (see Materials and Methods). HTS data were first analyzed for two independently sorted samples cultured in YPD medium. As shown in Fig. 3(A), the identified promoters within the two replicates are highly correlated ($R^2 = 0.92$), suggesting our sorting method is reproducible. To further confirm that promoters with different activity could be distinguished by FACS, 12 promoters from the most active fraction, i.e., Bin 12, were selected and re-tested individually. As shown in Fig. 3(B), 10 out of the 12 promoters show strong activity. Furthermore, analyzing these strong promoters shows that many of them drive the expression of ribosome proteins. This is reasonable because the ribosome proteins are among the highest expressed proteins in a cell. In addition, we also identified the known highly active promoter, pYGR192C (pTDH3) in the population. Therefore, the FACS analysis followed by HTS could be used to efficiently identify promoters from the library under the condition of interest. Future work should apply this method to different growth conditions and strains, especially those with high potential in industrial application [21–23].

In addition to high-throughput sequencing and to

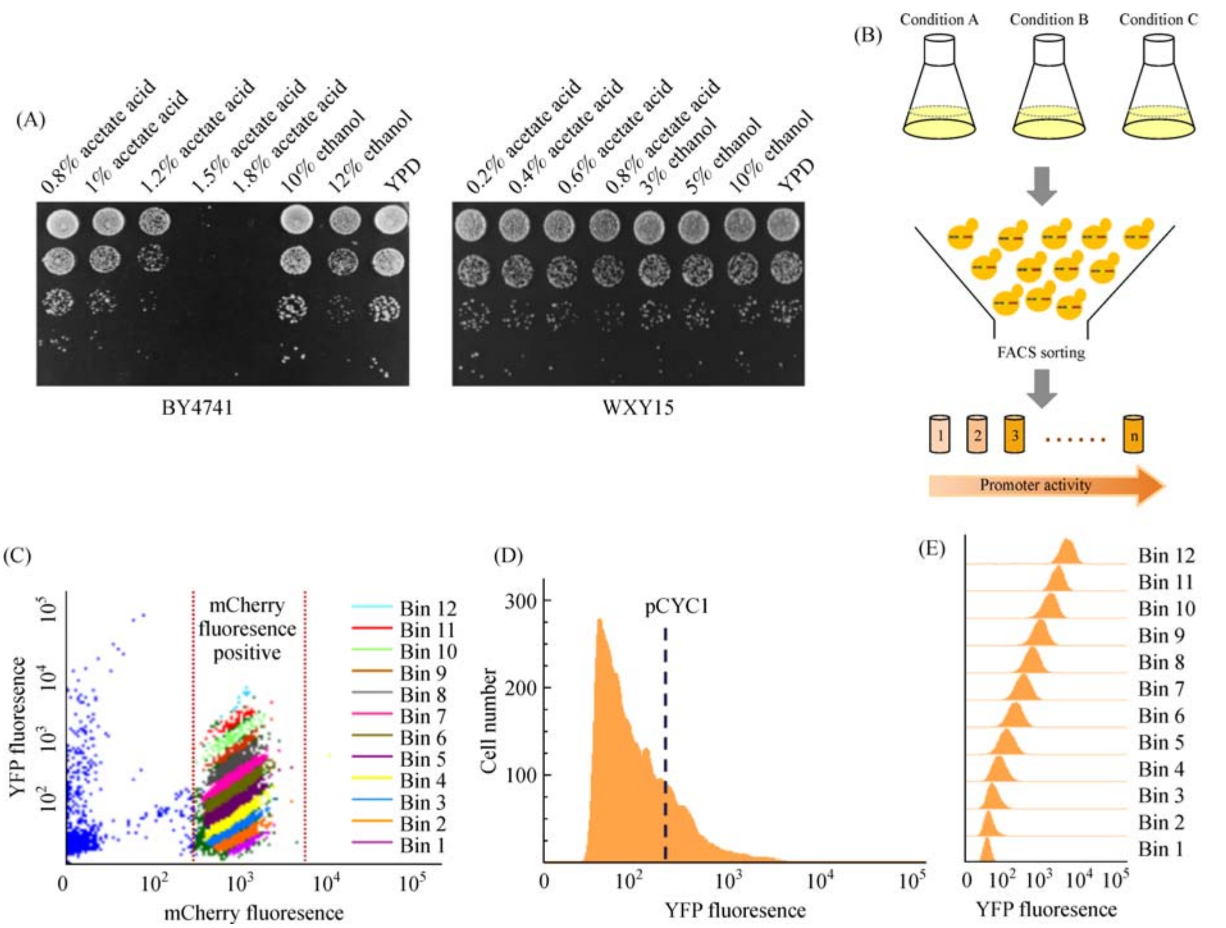


Fig. 2 Accurate expression measurement for all yeast native promoters under different growth conditions. (A) 10-fold serial dilution was used to determine the suitable stress conditions. BY4741 (left) and WXY15 (right) were treated with various concentration of acetic acid or ethanol, normalized to the starting $OD_{600} = 0.2$, 10-fold diluted and spotted onto YPD medium. Pictures were taken after the plates were incubated at 30 °C for two days; (B) illustration of the cell sorting experiment; (C) fluorescence plot of yeast library in WXY15 grown in YPD medium as an example. The dots between the two red dash lines represent cells expressing mCherry, which was constitutively expressed once the reporter fragment was integrated. Different expression bins are labeled by different color. The cells were divided into 12 bins, with increased expression from 1 to 12; (D) the distribution of YFP fluorescence in WXY15 library grown in YPD as an example. The blue dash line indicates the fluorescence intensive if YFP is under the control of CYC1 promoter; (E) the distribution of YFP fluorescence of cells from the sorted bins. The sorted cells from each expression bin were cultured and analyzed separately

quickly identify the promoters, four clones were also randomly selected from each of the three strongest Bins (10, 11 and 12), and subjected to FACS analysis to confirm the promoter activity. As shown in Fig. 3(C), clones from each bin show gradually increased activity from Bin 10 to Bin 12, which is consistent with the measurement of the entire population in Fig. 2(E). The promoter in each clone was then determined by PCR followed by Sanger sequencing (Table 3). We identified several promoters, which were commonly found as the strongest ones (in Bin 12) in both strain backgrounds and various growth conditions. This result was further confirmed from Figs. 3(D) and 3(E) in BY4741 and WXY15, respectively). These results show that pHSP26 is largely induced by ethanol in WXY15 but not in BY4741, whereas pYBR118W maintains high activity under ethanol stress in WXY15 but not in BY4741. In both strains, pYHR174W, pYKL060C and pYKL096W-A were strong promoters across all growth conditions. Upon acetic acid treatment, the fluorescence ratio of all promoters was reduced in both strains, although the level of reduction differs in different strain, suggesting that same promoter might function differently in different genetic background. Comparing with the activity of these promoters with other commonly used promoters (pPGK1, pTEF1, pADH1, pTDH3 promoters) in metabolic engineering, the promoters identified here could behave even better (Fig. 3(D)). Therefore, highly active promoters could be picked out by both high-throughput sequencing and single colony identification. The bacteria and yeast library presented in this study could provide the community a tool to quickly identify promoters with desired activity under a defined condition of interest in your favorite strain.

Previously, much work has been done on promoter

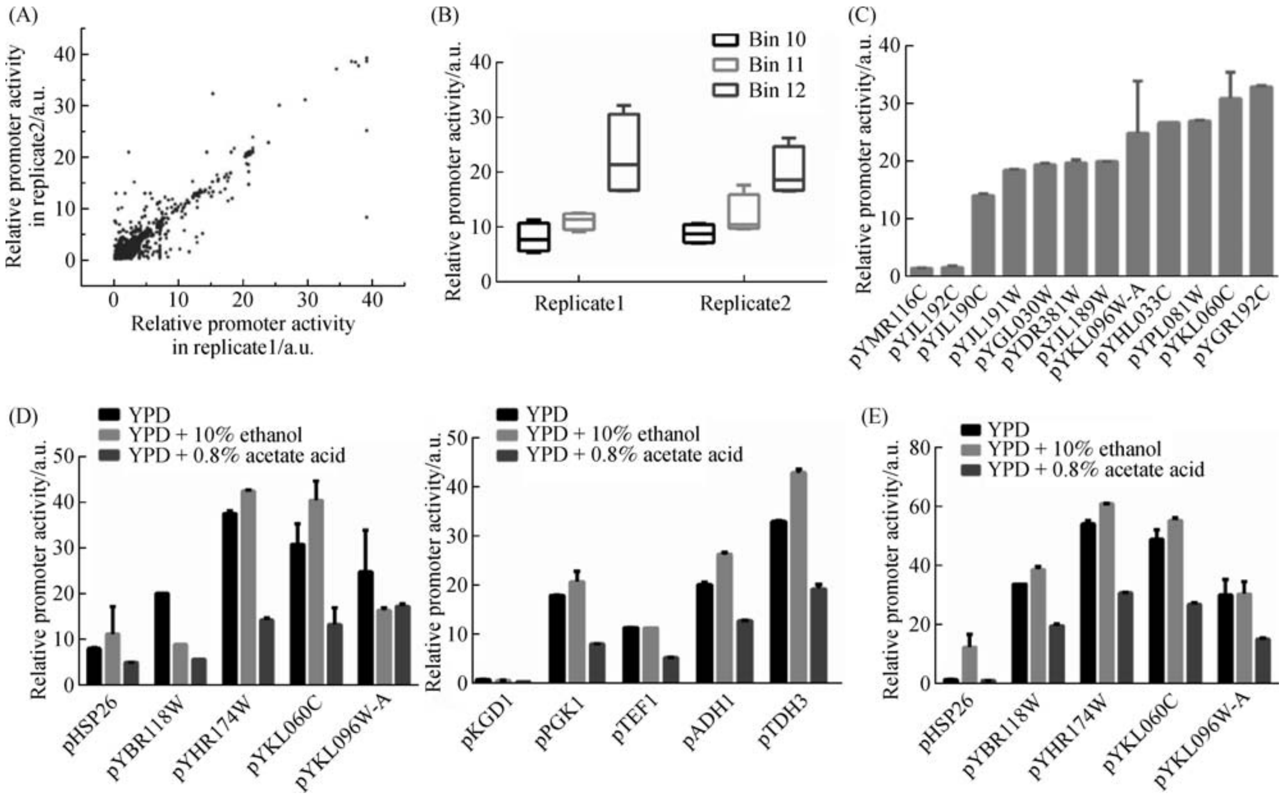


Fig. 3 Identify promoters with high activity under stress condition. (A) Comparison of the expression measurements obtained for two independent replicates (cultivated in YPD). For each replicate, the cells were sorted by FACS and subjected to high-throughput sequencing to identify the promoter in sorted expression bins. a.u., arbitrary unit; (B) confirm the activity of identify promoters from NGS sequencing. Independent clones were analyzed by FACS. Five promoters that frequently identified in almost all conditions from Bin 12 were examined individually in different conditions in BY4741 (D, left panel) or WXY15 (F), respectively. In BY4741, the five selected promoters (left panel) generally performed even better than some commonly used promoters (right panel); (C) expression measurement of randomly isolated clones from selected expression bins in two independent replicates. At least four clones were tested for each bin

mutagenesis or synthetic promoters [6,24,25]. However, only a limited number of promoters was highly active and they are repetitively used to construct a metabolic pathway to control the expression of multiple genes. These repetitive sequences lead to difficulty in pathway construction [7]. Besides, short synthetic promoters often consists of a few core elements [26], and their design and characterization are usually based on laboratory strain or one targeted stress condition [27], thus the flexibility is very limited. In contrast, our library contains thousands of yeast native promoters throughout the yeast genome. Using FACS-based screening method, it could, in theory, provide many desired promoters at any given conditions for industrial application.

Table 3 Promoters identified from random selected clones within Bin 12

Yeast strain	YPD	Ethanol stress	Acetic acid stress
BY4741	pYKL081W	pYJL159W	pYJL59W
	pYPL081W	pYPL081W	pYHR174W
	pYGL030W	pYHR174W	pYFR049W
	pYJL159W	pYKL096W-A	pYDL229W
	pYHR174W	pYJR105W	pYBR123C
	pYFR049W	pYBR072W	pYBR118W
			pYBR072W
WXY15	pYDR381W	pYGL030W	pYKL060C
	pYBR072W	pYBR118W	
	pYGL030W	pYHR174W	
	pYJL159W	pYPR145W	
	pYMR116C	pYDR381W	
		pYPL081W	

3.3 Improve xylose fermentation by constructing the utilization pathway with selected promoters

To test if the promoters identified in above strategy could be applied in industrial strain construct, we introduced the xylose metabolic pathway in BY4741. For yeast to metabolite xylose, xylose has to be converted to xylulose at first. The isomerization of xylose to xylulose is catalyzed by the xylose isomerase (XI) [28,29]. The XI pathway eliminates the cofactor imbalance and accompanying excessive production of xylitol. Xylulose is subsequently phosphorylated to D-xylulose-S-phophate (XSP) by xylulokinase (XK), and channeled through the pentose phosphate pathway (PPP) to glycolysis [30] (Fig. 4(A)).

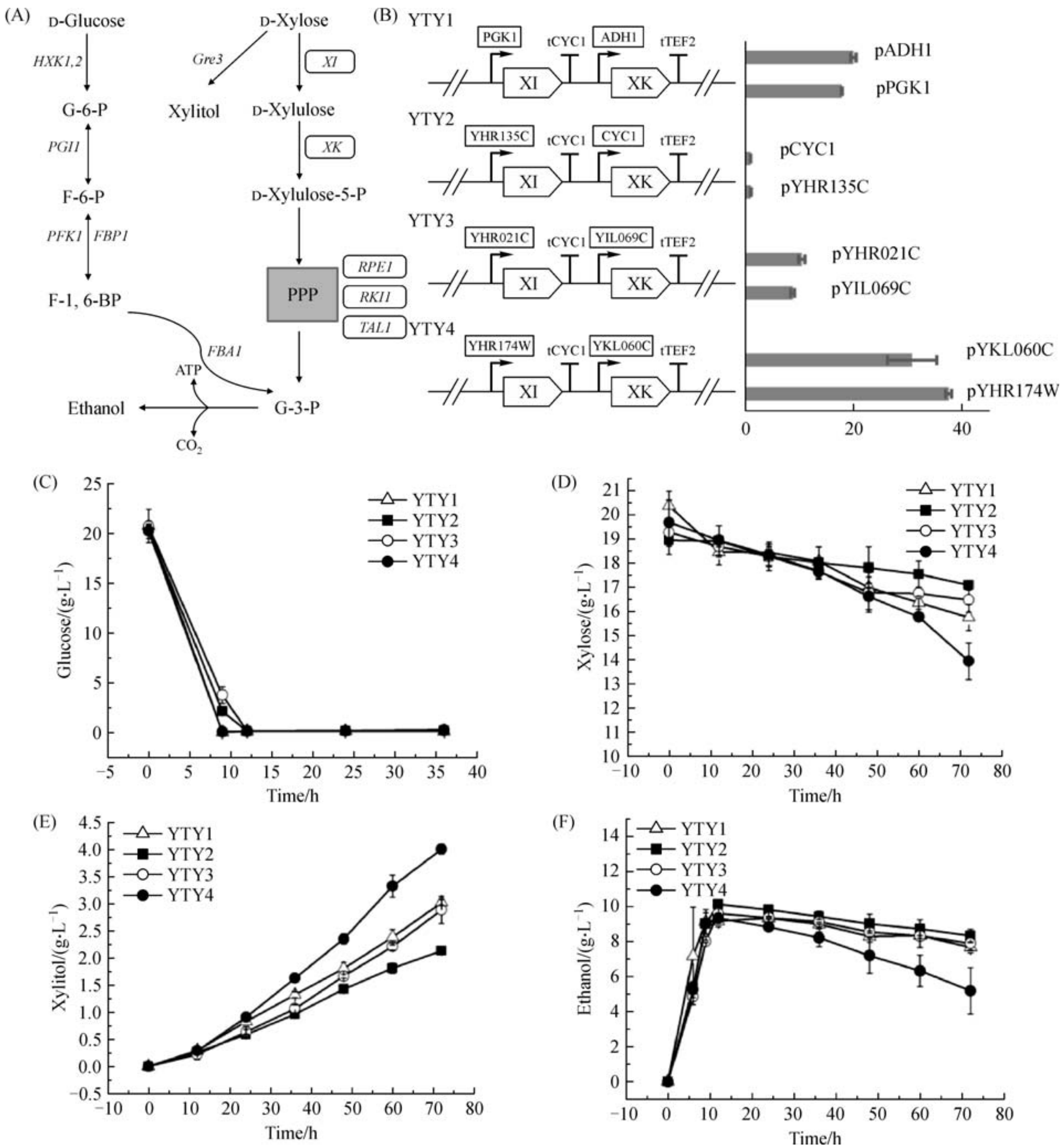


Fig. 4 Construction and characterization of xylose-fermenting yeast using identified promoters. (A) An overall scheme for glucose and xylose metabolism in engineered *Saccharomyces cerevisiae*. G-6-P, glucose-6-phosphate; F-6-P, fructose-6-phosphate; F-1,6-BP, fructose-1,6-biphosphate; G-3-P, glyceraldehyde-3-phosphate; PPP, pentose phosphate pathway. To simplify the analysis, only the expression of genes in rectangle was modulated; (B) the yeast strains with XI and XK under the control of promoters with different activities; (C) the residual concentration of glucose during fermentation; (D) the residual concentration of xylose during fermentation; (E) the productivity of xylitol during fermentation; (F) the productivity of ethanol during fermentation

Therefore, we targeted XI and XK to optimize their expression.

XI and XK were assembled into transcription units (TUs) by varying the promoters and/or terminators, and both TUs were integrated into the yeast genome following the Yeastfab method as reported previously [7]. For comparison, the commonly used strong promoters, pPGK1 and pADH1, were employed to drive XI and XK expression, respectively. Eventually, four strains were generated including both XI and XK under the control of weak (YTY2), medium (YTY3), strong (YTY4) and common promoters (YTY1) (Table 2, Fig. 4(B)). The four yeast strains were subjected to fermentation in medium containing 20 g/L of glucose and 20 g/L of xylose.

As shown in Fig. 4(C), glucose was used out within 12 h by each strain, suggesting that all strains could utilize glucose similarly. Once glucose was consumed completely, the strains switched to use xylose as carbon source. By monitoring the amount of xylose in the media, we found the rate of xylose consumption is correlated with the activity of promoters. The stronger the promoters are, the faster xylose is consumed. YTY4, in which both genes were under the control of the strong promoters, pYKL060C and pYHR174W, could utilize xylose faster than the control strain, suggesting better promoter could be identified and utilized under this particular cultivation condition.

During fermentation, the yeast strains also produced xylitol and ethanol. As shown in Fig. 4(E), the yield of xylitol in the four strain increases with the increase of the promoter activity, similar to above mentioned trend on consumption of xylose. However, as shown in Fig. 4(F), the tendency of ethanol production was different. YTY4 had the lowest ethanol productivity. It was reported that xylitol in the medium could inhibit the production of ethanol in a concentration dependent manner, and *GRE3* deletion could reduce the byproduct xylitol [31]. This could explain why strain YTY4 with active *GRE3* produces higher xylitol but lower ethanol. These results suggested that the selected promoters could improve xylose utilization than the one with common promoters reported previously [13]. Therefore, our library together with FACS analysis could be useful in finding proper promoters for metabolic engineering.

4 Conclusions

We have constructed a collection of genome-wide promoter libraries (including 9 bacteria and 2 yeast libraries) with each promoter cloned into a reporter plasmid. In combination with FACS followed by high-throughput sequencing analysis, we demonstrated that promoters with different activities could be separated and efficiently identified under a given condition. Several promoters showing strong activity under both normal and stress conditions were used to express key enzymes in xylose fermentation pathway, resulting in improved xylose utilization. Therefore, these libraries could provide a useful tool to identify desired promoters to optimize the metabolic pathways.

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